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# Cyber Security Risk Assessment Spring 2025

*Lecture 07: Identifying Primary and Supporting Asset EXERCISE REVIEW*

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# Grade Score

**Well done**

**You can do better**

**No ..**

# Primary Assets

| | |
|---|---|
| **Case Study** | *Remotely piloted fleet* |

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# Primary Assets

| Step 1: Impact Assessment | | |
|---|---|---|
| 1.1. Primary Asset (PA) Identification | | |
| *Primary Asset ID* | *Primary Asset Name* | *Type (information/service)* |
| 1 | *Send signals to other drones* | service |
| 2 | *Provide weather conditions* | service |
| 3 | *Provide information about the air traffic* | service |
| | | |
| 4 | *Information about the surrounded environment and the land environment* | information |
| 5 | *Provide authentication of the drone* | service |

- **Power**
- **Monitoring**
- **Connections**

# Impact Assessment

| Step 1: Impact Assessment | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| 1.2 Impact Assessment on Primary Assets | | | | | | | | | | |
| *Primary Asset Name* | *Potential Compromise of C, I* | *Impact (see Table in the Methodology)* | | | | | | | | |
| | | Personnel | Capacity | Performance | Economic | Branding | Regulatory | Environment | Overall Impact (1,2) | Justification |
| 1 | C | 1 | 1 | 1 | 2 | 1 | 2 | 1 | 2 | if the information are disclosured the main problem for the company is that it can have an economic problem but the other conditions are safe |
| | I | 1 | 3 | 3 | 3 | 2 | 3 | 2 | 3 | if the information are modified by someone the drones have the wrong information about the positions of the other drones and this can cause a huge amount of problems in terms of capacity or perfomances |
| | A | 1 | 3 | 3 | 3 | 2 | 2 | 3 | 3 | if the information are unavailable the drones cannot do nothing, the valuation of integrity and availability is the same but for different justification because in this case the impact into the environment could be more dangerous than the previous one |
| 2 | C | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | there isn't a huge impact of this service if there's a disclosure of information, so I think the impact is 1 in all the parameters that need to be considered. |
| | I | 1 | 4 | 4 | 3 | 1 | 2 | 2 | 4 | both in Integrity and availability the impact is the same because they can cause problems if the meteo data are unavailable or wrong, the parameter that is different is the capacity because if the data aren't available the drone can come back at its hub but if they are wrong the drone can have the wrong behaviour and we don't know the consequences on it |
| | A | 1 | 3 | 4 | 3 | 1 | 2 | 2 | 4 | |
| 3 | C | 1 | 1 | 1 | 2 | 1 | 2 | 1 | 2 | i think that the disclosure of information it's not too dangerous because they are available to a lot of companies that manage the airspace but I think that the disclosure of privacy information about what type of vehicles are in the airspace could cause economic problems to the company |
| | I | 1 | 2 | 3 | 4 | 4 | 3 | 3 | 4 | the compromisation of traffic information can attack the way in which the drones are flying and this has as a consequences the crash between them and this is why the overall impact is considered 4 |
| | A | 1 | 4 | 4 | 4 | 4 | 4 | 3 | 4 | from availability point of view even if the damage could be minor the most important problem is related to the performance and the capacity, because if a drone can't have the information related to the traffic it cannot be on air and it is force to land in the hub, so the performance and the capacity have more impact than the other parameters into the overall calculation |

# Supporting Assets

| Step 2: Supporting Asset Identification & Valuation | | | | | |
|---|---|---|---|---|---|
| 2 Linkage with Primary Assets | | | | | |
| ***Primary Asset / Supporting Asset*** | Send signals to other drones | Provide weather conditions | Provide information about the air traffic | Information about the surrounded environment and the land | Provide authentication of the drone |
| *Sensors for radio frequency signals* | X | | | | |
| *GPS* | X | X | X | X | |
| *Sensors for meteo conditions* | | X | | | |
| *Radar* | | | X | X | |
| *Firewall* | | | | | X |
| *Database* | | | | | X |
| *MFA authentication* | | | | | X |
| *Microcamera* | | | | X | |
| | | | | | |
| *Router* | X | | | | |
| | | | | | |

# Primary Assets

| Step 1: Impact Assessment | | | |
|---|---|---|---|
| 1.1. Primary Asset (PA) Identification | | | |
| ***Primary Asset ID*** | ***Primary Asset Name*** | ***Type (information/service)*** | ***Justification*** |
| 1 | *deliver goods* | service | main objective of this kind of UTS |
| 2 | *safety among vehicals* | service | avoid incidents ruining the goods and the vehicles |
| 3 | *sensitive information* | information | avoid the disclosure or loss of information related to the delivery (e.g customer, goods) |
| 4 | *connection with the station* | service | comprimising it could cause the loss of the track and not knowing where to fly |
| 5 | *customer list* | information | important infomation for the business scope |
| 6 | *congestion control algorithm* | information | how to avoid incidents |

- **Power**
- **Monitoring**
- **Connections**

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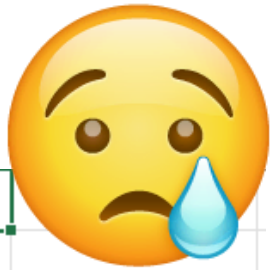
# Impact Assessment

**Step 1: Impact Assessment**

**1.2 Impact Assessment on Primary Assets**

| *Primary Asset Name* | *Potential Compromise of C, I or A:* | *Impact (see Table in the Methodology)* | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | **Personnel** | **Capacity** | **Performance** | **Economic** | **Branding** | **Regulatory** | **Environment** | **Overall Impact (1.2)** | **Justification** |
| deliver goods | C | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | It has no impact on confidentiality |
| | I | 1 | 2 | 2 | 1 | 1 | 1 | 1 | 2 | If the system is compromised there would be integrity problem |
| | A | 1 | 3 | 3 | 3 | 3 | 1 | 1 | 3 | High impact on availability and branding/reputation |
| safety among vehicles | C | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | No problems with condifentiality |
| | I | 1 | 5 | 5 | 4 | 4 | 1 | 2 | 5 | The system is compromised in terms of integrity, potentially many vehicles demaged |
| | A | 1 | 5 | 5 | 4 | 4 | 1 | 1 | 5 | Same as integrity, the system will not be available |
| sensitive information | C | 1 | 1 | 1 | 4 | 4 | 5 | 1 | 4 | Confidentiality loss for spreading information, economic loss, reputation and law problems |
| | I | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | No problems in terms of integrity |
| | A | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | No problems in case of availability |
| *connection with the station* | C | 1 | 3 | 3 | 3 | 3 | 3 | 1 | 3 | Potential disclosure of important information about the routes |
| | I | 1 | 3 | 3 | 3 | 3 | 1 | 1 | 3 | If the attacker is able to compromise the route casuing a change in it, there is a loss of integrity |
| | A | 1 | 3 | 3 | 3 | 3 | 1 | 1 | 3 | The system availability could be compromised |
| *customer list* | C | 1 | 1 | 1 | 4 | 4 | 5 | 1 | 4 | Sensitive information disclosed |
| | I | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | |
| | A | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | |
| *congestion control algorithm* | C | 1 | 1 | 1 | 4 | 4 | 1 | 1 | 3 | Potential loss in reputation and economic gain for copied of the algorithm |
| | I | 1 | 3 | 3 | 3 | 3 | 1 | 1 | 3 | If changed, problems with the integrity of the system |
| | A | 1 | 3 | 3 | 3 | 3 | 1 | 1 | 3 | If changed maliciously, potential problems with system availability |

# Supporting Assets



| Step 2: Supporting Asset Identification & Valuation | | | | | | |
|---|---|---|---|---|---|---|
| 2 Linkage with Primary Assets | | | | | | |
| ***Primary Asset / Supporting Asset*** | deliver goods | safety among vehicles | sensitive information | connection with the station | customer list | congestion control algorithm |
| *fleet* | x | | | | | |
| *incidents control* | | x | | | | x |
| *access control system* | x | x | x | x | x | x |
| *protection mechanisms* | x | | x | x | x | |
| *antennas for receiving commands* | | x | | x | | |
| | | | | | | |
| | | | | | | |
| | | | | | | |
| | | | | | | |
| | | | | | | |
| | | | | | | |

# Primary Assets

- Power
- Monitoring
- Connections

| Step 1: Impact Assessment | | | |
|---|---|---|---|
| 1.1. Primary Asset (PA) Identification | | | |
| ***Primary Asset ID*** | ***Primary Asset Name*** | ***Type (information/service)*** | ***Justification*** |
| 1 | *UTM Authentication System* | service | it's a service that provides authentication and access to the UTM system |
| 2 | *Airspace Design & Geo-fencing* | service | it's a service that occupies with defining an airspace and checking the restrictions |
| 3 | *Congestion Prediction* | information | it has all of the information and analysis about the predictions of the airspace congestion |
| 4 | *Separation Management* | service | a service that occupies with the safe separation of the aircrafts without pilots |
| 5 | *Communication Protocols* | information | it is a set of structured information regarding the communication rules between the UTM and UAS |

# Impact Assessment

**Step 1: Impact Assessment**

**1.2 Impact Assessment on Primary Assets**

| *Primary Asset Name* | *Potential Compromise of C, I or A:* | *Impact (see Table in the Methodology)* | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | **Personnel** | **Capacity** | **Performance** | **Economic** | **Branding** | **Regulatory** | **Environment** | **Overall Impact (1.2)** | **Justification** |
| UTM Authentication System | C | 1 | 2 | 3 | 3 | 5 | 5 | 1 | 5 | unauthorized people can access the system if the data are compromised |
| | I | 1 | 3 | 4 | 4 | 4 | 5 | 1 | 5 | errors in data can create problems in the system such as blocks or unauthorized access |
| | A | 2 | 5 | 5 | 4 | 3 | 5 | 3 | 5 | users can't access anymore the UTM functionalities |
| Airspace Design & Geo-fencing | C | 1 | 3 | 4 | 3 | 4 | 5 | 3 | 5 | the information about the airspace in the wrong hands can promote malicious attacks |
| | I | 1 | 4 | 5 | 3 | 4 | 5 | 3 | 5 | if data are inconsistent and modified, security issues and collisions can happen |
| | A | 2 | 5 | 5 | 4 | 3 | 5 | 3 | 5 | if the service is not accessible, managing the airspace can become caothic and unsafe |
| Congestion Prediction | C | 1 | 2 | 3 | 2 | 2 | 3 | 1 | 3 | stolen data can be used to manipulate traffic |
| | I | 1 | 3 | 5 | 4 | 3 | 4 | 2 | 5 | if the data are modified, accidents and delays can happen |
| | A | 1 | 4 | 5 | 4 | 3 | 4 | 2 | 5 | if the system is down, the traffic can't be managed, leading to dangerous congestions |
| Separation Management | C | 5 | 4 | 3 | 4 | 3 | 4 | 3 | 5 | if the data are compromised, the hackers can cause intentional collisions |
| | I | 5 | 5 | 5 | 5 | 5 | 5 | 4 | 5 | many accidents can happen if the integrity of the data is not guaranteed anymore |
| | A | 5 | 5 | 5 | 5 | 5 | 5 | 4 | 5 | similar to other assets, if the system is down the collision risk grows exponentially |
| Communication Protocols | C | 3 | 2 | 4 | 3 | 5 | 5 | 1 | 5 | if the communications are not safe, malicious users can intercept them and use the data for attacks |
| | I | 3 | 3 | 5 | 4 | 4 | 5 | 1 | 5 | modifying the protocols can lead to potential fake and inconsitent messages |
| | A | 3 | 4 | 5 | 4 | 4 | 5 | 1 | 5 | systems can't coordinate if the communication network is offline |

# Supporting Assets

**Step 2: Supporting Asset Identification & Valuation**

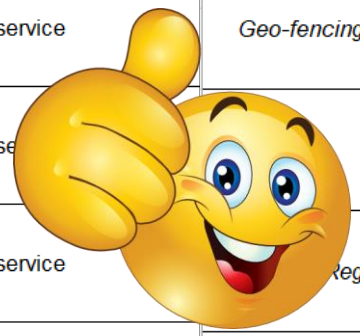
**2 Linkage with Primary Assets**

| ***Primary Asset / Supporting Asset*** | UTM Authentication System | Airspace Design & Geo-fencing | Congestion Prediction | Separation Management | Communication Protocols |
|---|---|---|---|---|---|
| *Radar Systems* | | x | x | x | |
| *Cloud Infrastructure* | x | x | x | | x |
| *Network Communication* | x | | | x | x |
| *Ground Control Stations* | x | x | | x | x |
| *Weather Monitoring Systems* | | x | x | | |

# Primary Assets

- **Power**
- **Monitoring**
- **LAN e Internet Connections**

| ***Primary Asset Name*** | ***Type (information/service)*** |
|---|---|
| *Meteorological Information* | information |
| *Air-to-Ground Communication* | service |
| *Ground-to-Ground Communication* | se |
| *Secure exchange of data* | service |
| *Flight Planning & trajectory changes to ensure safety* | service |
| *Data Collection & Data Distribution* | service |
| *Coordination & Safety in Emergency Response* | service |



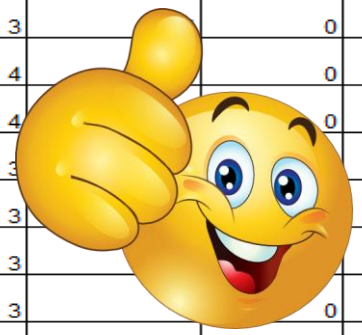
| | |
|---|---|
| *Syncronization with different service providers* | service |
| *Geo-fencing design and dynamic adjustments* | service |
| *congestion prediction* | service |
| *Registration & Authentication* | service |
| *Traffic, Aircraft & Unmanned information* | information |
| *Managing transition locations and locations with NAS* | service |
| *3D-Maps information* | information |
| *Airspace Constraint information* | information |

# Impact Assessment

**Step 1: Impact Assessment**

**1.2 Impact Assessment on Primary Assets**

| *Primary Asset Name* | *Potential Compromise of C, I or A:* | *Impact (see Table in the Methodology)* | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | Personnel | Capacity | Performance | Economic | Branding | Regulatory | Environment | Overall Impact (1.2) |
| Meteorological Information | C | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| | I | 0 | 3 | 2 | 0 | 3 | 3 | 0 | 3 |
| | A | 0 | 3 | 4 | 0 | 3 | 3 | 0 | 4 |
| Air-to-Ground Communication | C | 0 | 3 | 4 | 0 | 3 | 4 | 0 | 4 |
| | I | 0 | 4 | 4 | 0 | 3 | 4 | 0 | 4 |
| | A | 0 | 4 | 5 | 0 | 4 | 4 | 0 | 5 |
| Ground-to-Ground Communication | C | 0 | 3 | [illegible] | 0 | 3 | 4 | 0 | 4 |
| | I | 0 | 4 | [illegible] | 0 | 3 | 4 | 0 | 4 |
| | A | 0 | 4 | [illegible] | 0 | 4 | 4 | 0 | 5 |
| Secure exchange of data | C | 0 | 3 | [illegible] | [illegible] | 3 | 4 | 0 | 4 |
| | I | 0 | 3 | [illegible] | [illegible] | 4 | 4 | 0 | 4 |
| | A | 0 | 3 | [illegible] | [illegible] | 4 | 4 | 0 | 4 |
| Flight Planning & trajectory changes to ensure safety | C | 0 | 3 | [illegible] | 0 | 2 | 3 | 0 | 4 |
| | I | 0 | 3 | 5 | 0 | 3 | 4 | 0 | 5 |
| | A | 0 | 4 | 5 | 0 | 3 | 4 | 0 | 5 |
| Data Collection & Data Distribution | C | 3 | 3 | 2 | 0 | 3 | 3 | 0 | 3 |
| | I | 3 | 2 | 3 | 0 | 2 | 3 | 0 | 3 |
| | A | 0 | 4 | 3 | 0 | 3 | 3 | 0 | 4 |
| Coordination & Safety in Emergency Response | C | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| | I | 3 | 2 | 3 | 0 | 0 | 4 | 0 | 4 |



# Supporting Assets

**Step 2: Supporting Asset Identification & Valuation**

**2 Linkage with Primary Assets**

| ***Primary Asset / Supporting Asset*** | Meteorological Information | Air-to-Ground Communication | Ground-to-Ground Communication | Secure exchange of data | Flight Planning & trajectory changes to ensure safety | Data Collection & Data Distribution | Coordination & Safety in Emergency Response |
|---|---|---|---|---|---|---|---|
| *Power infrastructure* | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| *Network routers and switches* | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| *Servers* | 0 | 1 | 1 | 1 | 1 | 1 | 1 |
| *Radar systems* | 0 | 1 | 0 | 0 | 1 | 1 | 1 |
| *Firewalls* | 1 | 0 | 0 | 1 | 1 | 1 | 1 |
| *Operating systems (Specific)* | 0 | 1 | 1 | 1 | 1 | 1 | 1 |
| *Humans* | 1 | 1 | 1 | 1 | 1 | 0 | 1 |
| *Workstations for personnel* | 0 | 1 | 1 | 1 | 1 | 1 | 1 |
| *Ground control stations* | 1 | 1 | 1 | 1 | 1 | 0 | 1 |
| *Cells* | 0 | 1 | 1 | 1 | 1 | 1 | 1 |
| *Satellites* | 0 | 1 | 1 | 1 | 1 | 1 | 1 |
| *Third Party API (Specific)* | 1 | 0 | 0 | 0 | 0 | 0 | 1 |

# Primary Assets

- Power
- Monitoring
- LAN e Internet Connections

| Step 1: Impact Assessment | | | |
|---|---|---|---|
| 1.1. Primary Asset (PA) Identification | | | |
| ***Primary Asset ID*** | ***Primary Asset Name*** | ***Type (information/service)*** | ***Justification*** |
| PA1 | *Knowledge* | Information | in terms of "know-how". If someone leaves the company and there is not documentation or it is not complete (bad but actyally in real world can happen), then it is possible to lose a lot |
| PA2 | *Sensitive Information, Da* | Information | This primary asset refers both to personal data (information about the workers) and both to system/architecture/infrastructure information and also company's data |
| PA3 | *Contingency management* | Service | will include but are not limited to emergency landing site guidance and geo-fencing segments of airspace due to wind to ensure safe operations. |
| PA4 | *Predicted wind and weather* | Service | |
| PA5 | *Real time weather and wind* | Service | |
| PA6 | *Flight plan trajectory* | Service | This is one of the main services in order to provide a safe trajectory, in particular in case of emergency |
| PA7 | *Airspace constraints (dynamically adjusted)* | Service | Dynamically adjusted, important service that needs to be reliable as it is a real-time service which indicates airspace constraints |
| PA8 | *Web Application/System* | Service | This refers to web application used in order to collect data |

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# Impact Assessment

**Step 1: Impact Assessment**

**1.2 Impact Assessment on Primary Assets**

| Primary Asset Name | Potential Compromise of C, I or A: | Impact (see Table in the Methodology) | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | Personnel | Capacity | Performance | Economic | Branding | Regulatory | Environment | Overall Impact (1-2) | Ju |
| Knowledge | C | 1 | 2 | 2 | 3 | 3 | 3 | 1 | 3 | If there is some critical pro confident, than it could affe |
| | I | 1 | 5 | 3 | 3 | 3 | 2 | 1 | 5 | If there is no more "kno the activities will b |
| | A | 1 | 5 | 4 | 4 | 3 | 1 | 1 | 5 | If there is no more "know-how", then the process and so the activities will be strongly affected about the |
| Sensitive Information, Data | C | 3 | 3 | 3 | 3 | 3 | 5 | 1 | 5 | E.g. If a data breach happen and personal data are stolen |
| | I | 1 | 4 | 4 | 3 | 3 | 5 | 1 | 5 | The change of sensitive information and data can affect the whole system seriously |
| | A | 2 | 5 | 5 | 4 | 3 | 4 | 2 | 5 | The non-availability of sensitive data could also lead to disaster |
| Contingency management | C | 1 | 3 | 1 | 3 | 3 | 3 | 1 | 3 | |
| | I | 1 | 5 | 5 | 3 | 4 | 5 | 2 | 5 | If there is the need to do an emergency landing due to something bad and this information is not available or is |
| | A | 1 | 4 | 4 | 3 | 3 | 3 | 1 | 4 | If there is the need to do an emergency landing due to something bad and this information is not available or is |
| Predicted wind and weather | C | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | It is not a confidential information |
| | I | 1 | 2 | 2 | 3 | 2 | 1 | 1 | 3 | Its integrity can also be damage but it won't affect that much the system. Maybe some flight won't departure |
| | A | 1 | 4 | 4 | 4 | 3 | 2 | 1 | 4 | The absence could lead to the stop of some services |
| Real time weather and wind | C | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | |
| | I | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | |
| | A | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | |
| Flight plan trajectory | C | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | |
| | I | 1 | 5 | 4 | 3 | 3 | 5 | 2 | 5 | The change of a flight trajectory could lead to a disaster |
| | A | 1 | 5 | 4 | 3 | 2 | 4 | 1 | 5 | The sudden unavailability of the flight plan trajectory could lead to disaster |
| Airspace constraints (dynamically adjusted) | C | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | |
| | I | 1 | 5 | 5 | 4 | 4 | 4 | 3 | 5 | If the constraints are modified and not the real one, then it could cause serious incidents |
| | A | 1 | 5 | 5 | 4 | 4 | 4 | 3 | 5 | E.g. emergency plan: if there is no dynamically adjustment of airspace constraints could lead to a disaster |
| Web Application/System | C | 1 | 3 | 4 | 3 | 4 | 4 | 1 | 5 | Exposure of information system can expose the who system to attacks |
| | I | 1 | 4 | 5 | 4 | 4 | 5 | 1 | 5 | The change in behavior of the system could lead to damage/interrupt services |
| | A | 1 | 4 | 4 | 4 | 3 | 4 | 2 | 5 | The unavailability of the system can lead to interruption of the entire service if no countermeasure are in place |

# Supporting Assets

**Step 2: Supporting Asset Identification & Valuation**

**2 Linkage with Primary Assets**

| ***Primary Asset / Supporting Asset*** | Knowledge | Sensitive Information, Data | Contingency management | Predicted wind and weat | Real time weather and wind | Flight plan trajectory | Airspace constraints (dynamically adjusted) | Web Application/System |
|---|---|---|---|---|---|---|---|---|
| *People* | x | x | | | | | | |
| *O.S. version (Fc, Server, DB)* | | x | | | | | | x |
| *Documentation* | | x | | | | | | |
| *Datacenter* | | | | | | | | x |
| *Low-altitude radar* | | | x | | | x | x | |
| *Cell* | | | x | | | | | |
| *Satellite* | | | x | x | x | | | |
| *three-dimensional map* | | | x | | | x | x | |

# Primary Assets

| Step 1: Impact Assessment | | | |
|---|---|---|---|
| 1.1. Primary Asset (PA) Identification | | | |
| ***Primary Asset ID*** | ***Primary Asset Name*** | ***Type (information/service)*** | ***Justification*** |
| PA1 | *link/connection between UAS and UTM* | service | The connection between the UAS and the UTM: foundamental asset to keep track of the UAS |
| PA2 | *authentication/detection of other UAS* | service | An important service is to detect, identify, authenticate and verify other UAS that share the same air space. |
| PA3 | *Geo-fencing / Airspace definition* | service/ information | Ability to use geo-spatial information to create safe trajectories. Primary asset that defines the accuracy and the safety of the UAS |
| PA4 | *CNS (communication, navigation and surveillance)* | Service | Provide communication, navigation and surveillance at any moment, regardless day time or weather conditions. |
| PA5 | *Prediction and separation* | Service | Ability to predict potential collisions between UAS o between other unidentified moving objects e.g. birds, wind turbines... |
| PA6 | *Weather monitoring* | information / service | Weather data is a primary asset used to carry on the service of routing the UAS away from adverse weather conditions. |
| PA7 | *Business confidential trajectories* | information | Specification needed in case the purpose of the flight needs to be confidential. The information about the confidential routes need to be protected. |

- **Power**
- **Monitoring**
- **LAN e Internet Connections**

# Impact Assessment

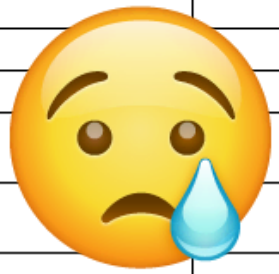
| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | Reverse engineering | No impact | No impact | Minor | Minor | Minor | No impact | No impact | Minor | A potential confidentiality breach could result in finding flaws in the technology used for the prediction system |
| PA5 | Feeding false data | No impact | Minor | Severe | Severe | Severe | Minor | Minor | Minor | The ability to feed false data to a prediction system could impair the knowledge on the UAS on what consitutes an airborne object to avoid. Even though a potential collision could be a severe problem, the difficulty of retraining a UAS in overseeing potenital collision is harder than the benefits it provides. The bigger problems concern performance, economics (e.g. loosing too many UAS due to collisions) and branding. |
| | Interrupting the prediction system. | No impact | Minor | Minor | Minor | Minor | No impact | No impact | Minor | The overall impact of impairing the airborne collision system is minor given the difficulty of the attack and the unlikelyhood of facing serious conditions. |
| | Stealing data | No impact | No impact | No impact | Minor | No impact | No impact | No impact | No impact | There could be some confidential data in the waether field but I can't see significant security threats, if not collecting |
| PA6 | Tampering the waether data | No impact | No impact | Severe | Minor | No impact | No impact | No impact | Minor | This could be mainly a performance and an economic problem, since the UAS could be re-routed in areas with adverse weather conditions that could impair both the UAS and its cargo. |
| | Interrupting data stream | No impact | No impact | Minor | Minor | No impact | No impact | No impact | No impact | Potential accidental routing to an adverse weather area - unlikely. |
| | Stealing data | No impact | No impact | Severe | Minor | Critical | Minor | No impact | Severe | The routes are confidential, a confidentiality breach could have overall severe consequences on the service. The biggest problem in my opinion is the way this breach could affect branding, since trajectory confidentiality is not a core property to make the UAS work but is an additional feature that might convince customers to choose this brand over |
| PA7 | Tampering with the trajectories data | No impact | No impact | Minor | Minor | No impact | No impact | No impact | Minor | Having false trajectories in the database could lead to minor problems, moslty reguarding performance an economics. |
| | Deleting trajectories data | No impact | No impact | No impact | Minor | No impact | No impact | No impact | No impact | Overall no impact, since the problem is likely to happen before the UAS starts its flight. |

# Supporting Assets

**Step 2: Supporting Asset Identification & Valuation**

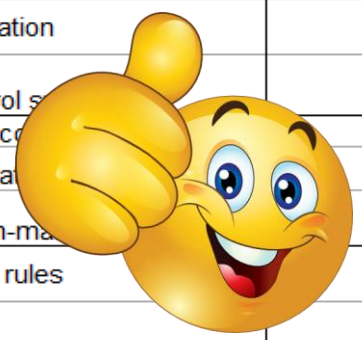
**2 Linkage with Primary Assets**

| ***Primary Asset / Supporting Asset*** | PA1 | PA2 | PA3 | PA4 | PA5 | PA6 |
|---|---|---|---|---|---|---|
| *Cellular networks (4G/5G)* | x | x | | x | | |
| *Satellite communication* | x | x | | x | | |
| *Identification system (e.g. ID reading)* | | x | | x | | |
| *Radar-based systems* | | x | | x | x | |
| *Geographic/Geospatial data models* | | | x | | x | x |
| *GPS* | | x | x | x | x | x |
| *Computer vision models* | | x | x | | x | |
| *Large scalable databases (e.g. Cassandra DB)* | | | x | | x | x |
| *Lidar systems* | | x | x | x | x | |
| *ML models for trajectory prediction* | | | | | x | |
| Weather stations | | | | | | x |



# Primary Assets

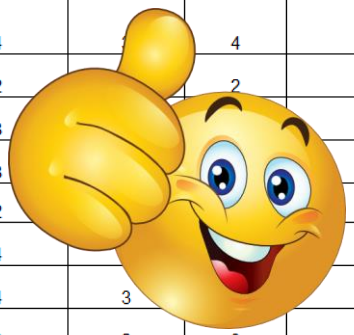
| Primary Asset ID | Primary Asset Name | Type (information/service) |
|---|---|---|
| P1 | User authentication & identity management | Service |
| P2 | Operator role permissions | Information |
| P3 | Drone monitoring & control | Service |
| P4 | Drone flight planning | Service |
| P5 | Real-time traffic management data | Information |
| P6 | Incidents detection alerts | Information |
| P7 | Encrypted flight telemetry data | Information |
| P8 | Secure communication channels | Service |
| P9 | Cross-domain authentication | Service |
| P10 | Inter-agency coordination | Service |
| P11 | Emergency override control s | Service |
| P12 | Mobile operator dashboards & c | Service |
| P13 | UTM and UAS operation da | Information |
| P14 | Automated UTM decision-ma | Service |
| P15 | Airspace geo-fencing rules | Information |
| P16 | Real time weather updates & forecast | Information |
| P17 | Weather impact prediction | Service |
| P18 | User Credentials | Information |
| P19 | Air traffic detection | Service |
| P20 | Inter facility UTM data exchange | Service |
| P21 | National airspace integration feed | Information |



- Power
- Monitoring
- LAN e Internet Connections

# Impact Assessment

| Primary Asset Name | Potential Compromise of C, I or A: | Impact (see Table in the Methodology) | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | Personnel | Capacity | Performance | Economic | Branding | Regulatory | Environment | Overall Impact (1.2) | Justification |
| User authentication & identity management | C | 3 | 2 | 3 | 3 | 4 | 4 | 1 | 4 | Significant impact on branding as well as violation of regulations |
| | I | 4 | 3 | 4 | 4 | 4 | 4 | 1 | 4 | If integrity is compromised, malicious actors could gain full access to UTM systems |
| | A | 3 | 4 | 4 | [illegible] | 4 | 3 | 1 | 4 | Auth dependent systems become inaccessible, causing severe performance degradation |
| Operator role permissions | C | 2 | 1 | 2 | [illegible] | 2 | 2 | 1 | 2 | exposed role permissions alone provide limited actionable capability without authentication credentials |
| | I | 4 | 3 | 3 | [illegible] | [illegible] | 3 | 2 | 4 | Altered permissions could enable unauthorised personnel to perform dangerous operations beyond their training or authorisation |
| | A | 3 | 3 | 3 | [illegible] | [illegible] | 2 | 1 | 3 | Temporary loss prevents role assignment, however it does not directly affect stop operations |
| Drone monitoring & control | C | 1 | 2 | 2 | [illegible] | [illegible] | 2 | 1 | 2 | Exposure of the service itself has limited impact, however attackers could gather useful information |
| | I | 5 | 4 | 4 | [illegible] | [illegible] | 5 | 3 | 5 | If attackers alter drone commands, this could cause crashes or unauthorised operations. |
| | A | 4 | 4 | 4 | 3 | [illegible] | 4 | 2 | 4 | If the service is unavailable, operators lose oversight of drones, posing severe risks. |
| Drone flight planning | C | 1 | 2 | 2 | 2 | 3 | 1 | 1 | 2 | Complaints about flight data security |
| | I | 4 | 3 | 3 | 3 | 4 | 4 | 2 | 4 | Corrupted flight plans could direct drones into restricted airspace or dangerous flight paths, potentially leading to multiple injuries |
| | A | 3 | 3 | 3 | 3 | 3 | 3 | 1 | 3 | Inability to create or access flight plans would severely disrupt drone operations, potentially leading to safety issues from ad-hoc unplanned flights |
| Real time traffic management data | C | 2 | 1 | 2 | 2 | 2 | 2 | 1 | 2 | While sensitive, traffic management data exposure alone provides limited actionable opportunities for malicious actors |
| | I | 5 | 4 | 4 | 4 | 5 | 5 | 3 | 5 | Manipulated data can cause drone mismanagement and regulatory failures. |
| | A | 4 | 4 | 4 | 3 | 4 | 4 | 2 | 4 | If unavailable, drone coordination becomes chaotic, leading to major disruptions. |



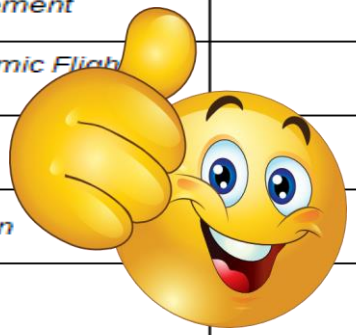
# Supporting Assets

| ***Primary Asset / Supporting Asset*** | User authentication & identity management |
|---|---|
| Servers | x |
| Identity management software | x |
| Databases | x |
| Multi-Factor authentication Devices | x |
| SIEM systems | |
| Workstations | x |
| RBAC Systems | x |
| Ground control stations | |
| Satellites | |
| Flight planning software | |
| Mission Control Software | |
| GIS mapping systems | |
| Firewalls | |
| Network access control systems | x |
| Intrusion detection systems | |
| Radar Systems | |
| Network Infrastructure | |
| Data processing units | |
| Security Operations Centre Systems | |
| Encryption modules | |
| Public Key Infrastructure Systems | x |

| |
|---|
| Air Traffic Management Software |
| Weather monitoring stations |
| Black Box Recorders |
| Cloud Computing infrastructure & Datacentres |
| VPNs |
| Electricity and Power Grids |
| Cooling systems |
| Physical Facilities (e.g. control rooms) |
| Highly skilled personnel |
| Workstation security policies |

# Primary Assets

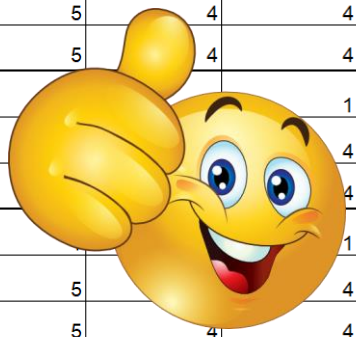
| Primary Asset ID | Primary Asset Name | Type (information/service) |
|---|---|---|
| PA1 | *Weather Monitoring and Meteorological Data* | information |
| PA2 | *Contingency management* | service |
| PA3 | *Air traffic* | information |
| PA4 | *Horizontal and vertical space* | information |
| PA5 | *Authentication* | information |
| PA6 | *Terrain and object avoidance* | information |
| PA7 | *Dynamically adjustement of geo-fencing areas* | information |
| PA8 | *Airspace Design and adjustement* | information |
| PA9 | *Trajectory's Prediction and Dynamic Flight Adjustmement* | service |
| PA10 | *Surveillance* | service |
| PA11 | *Persisten communication* | service |
| PA12 | *Secure communication* | service |
| PA13 | *Maintaing safe separation* | service |
| PA14 | *Congestion management* | service |
| PA15 | *Security* | service |



- **Power**
- **Monitoring**
- **LAN e Internet Connections**

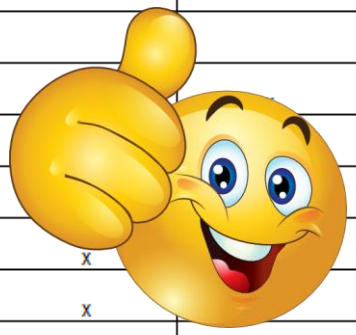
# Impact Assessment

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Secure communication | C | 1 | 1 | 1 | 3 | 4 | 5 | 1 | 4 | It could lead to exposure of sensible data like address |
| | I | 1 | 5 | 5 | 4 | 4 | 1 | 1 | 5 | By breaking the integrity of the communication the UAVs do not work anymore and they could be controlled changing the |
| | A | 1 | 5 | 5 | 4 | 4 | 1 | 1 | 5 | If not available the UAVs cannot fly |
| Maintaing safe separation | C | 1 | 1 | [illegible] | [illegible] | 1 | 1 | 1 | 1 | There is no probelm if someone know how the safe space is mantained |
| | I | 1 | 5 | [illegible] | [illegible] | 4 | 1 | 4 | 5 | Someone can modify it causing collision of UAVs to object or people |
| | A | 1 | 5 | [illegible] | [illegible] | 4 | 1 | 1 | 5 | If not available the UAVs cannot fly |
| Congestion management | C | 1 | 1 | [illegible] | [illegible] | 1 | 1 | 1 | 1 | There is no problem if someone knwon how the spacing and congestion is managed |
| | I | 1 | 5 | 5 | [illegible] | 4 | 1 | 3 | 5 | If the spacing is changed it could lead to collision between UAVs and cause damage |
| | A | 1 | 5 | 5 | 4 | 4 | 1 | 1 | 5 | If not available the UAVs cannot fly |
| Security | C | 1 | 5 | 5 | 5 | 5 | 1 | 1 | 5 | If someone know the security procedure it could compromise the system |
| | I | 1 | 5 | 5 | 5 | 5 | 1 | 1 | 5 | If someone modify the system it could lead to severe malfunctioning |
| | A | 1 | 5 | 5 | 5 | 5 | 1 | 1 | 5 | If not available the UTM cannot operate |



# Supporting Assets

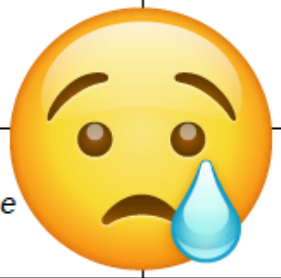
| ***Primary Asset / Supporting Asset*** | Weather Monitoring and Meteorological Data | Contingency management | Air traffic | Horizontal and vertical space | Authentication | Terrain and object avoidance | Dynamically adjustement of geo-fencing areas | Airspace Design and adjustement | Trajectory's Prediction and Dynamic Flight Adjustmement |
|---|---|---|---|---|---|---|---|---|---|
| *Personnel* | x | x | x | x | x | x | x | x | x |
| *Motors* | | x | | | | | | | x |
| *GPS* | | x | | | | | | | |
| *Cryptography* | | x | x | | x | | x | x | x |
| *Accelerometer* | | x | | | | | | | x |
| *Network communication between UAV and UTM* | | x | x | x | x | x | x | x | x |
| *Weather station and integration* | x | | | | | | | | x |
| *External air traffic data* | | x | x | x | | x | | x | x |
| *Terrain and Object Maps* | | | | | | x | x | x | x |
| *Proximity sensor* | | x | | | | | | | x |
| Antennas | | | x | | | | x | | |
| Database | x | x | x | | | x | x | x | x |
| UAV software | | x | | | | x | | | x |
| UTM software | x | x | x | x | | x | x | x | x |
| UTM hardware | x | x | x | x | | x | x | x | x |
| UAV hardware | | x | | | x | x | | | x |
| Electricity | x | x | x | x | x | x | x | x | x |
| UAV data storage | | x | | | x | x | | | x |
| UAV | | | | | x | | | | x |



# Primary Assets

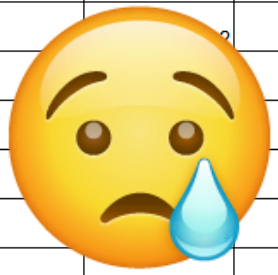
- Power
- Monitoring
- LAN e Internet Connections

| Step 1: Impact Assessment | | | |
|---|---|---|---|
| 1.1. Primary Asset (PA) Identification | | | |
| ***Primary Asset ID*** | ***Primary Asset Name*** | ***Type (information/service)*** | ***Justification*** |
| 1 | *Drone or Plane* | Service | Drone will be controled by an software |
| 2 | *Communication Network* | Information | wireless connection is needed to control the drone |
| 3 | *Software which controls the drone* | Service | Software will help in giving the comands to drone |
| | | | |
| 4 | *Camera* | Information | live camera feed help to give the direction |
| 5 | *Gps location* | Information | loaction can be tracked |



# Impact Assessment

| 1.2 Impact Assessment on Primary Assets | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| ***Primary Asset Name*** | ***Potential Compromise of C, I or A:*** | ***Impact (see Table in the Methodology)*** | | | | | | | | |
| | | **Personnel** | **Capacity** | **Performance** | **Economic** | **Branding** | **Regulatory** | **Environment** | **Overall Impact (1.2)** | **Justification** |
| Drone or Plane | C | 2 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | People Know about drones |
| | I | 3 | 2 | 3 | 3 | 1 | 3 | 3 | 3 | Comand givinen should be not tempered |
| | A | 2 | 1 | 2 | 3 | 3 | 3 | 3 | 3 | Should be availabe when needed |
| Communication Network | C | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | People Know about drones |
| | I | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | Connection should be secure |
| | A | 2 | 2 | 2 | 2 | [illegible] | [illegible] | 2 | 2 | Connection should always be available |
| Software which controls the drone | C | 1 | 1 | 3 | 1 | [illegible] | [illegible] | 3 | 3 | People Know about drones |
| | I | 3 | 2 | 3 | 2 | [illegible] | [illegible] | 4 | 4 | Cannot be hacked as is very dangerous |
| | A | 2 | 1 | 2 | 2 | [illegible] | [illegible] | 2 | 2 | software should always works |
| *Camera* | C | 4 | 4 | 4 | 4 | [illegible] | [illegible] | 4 | 4 | People Know about cameras in drones |
| | I | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | Should not be leaked |
| | A | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | if some time not availabe can also works |
| *Gps loaction* | C | 1 | 1 | 3 | 1 | 3 | 1 | 3 | 3 | People Know about drones |
| | I | 3 | 2 | 3 | 2 | 1 | 4 | 4 | 4 | Should not be traced |
| | A | 2 | 1 | 2 | 2 | 1 | 2 | 2 | 2 | if some time not availabe can also works |



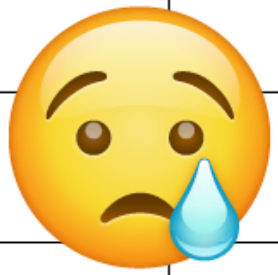
# Supporting Assets

| ***Primary Asset / Supporting Asset*** | Drone or Plane | Communi Network | Software which controls the drone | Camera | Gps loaction |
|---|---|---|---|---|---|
| *Storage for the camera feed* | x | | x | x | x |
| *ability to shoot* | x | | x | x | x |
| | | | | | |

# Primary Assets

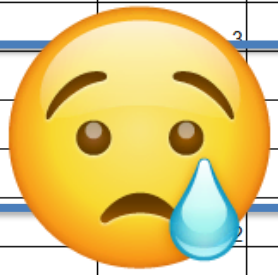
- **Power**
- **Monitoring**
- **LAN e Internet Connections**

| 1.1. Primary Asset (PA) Identification | | | |
|---|---|---|---|
| ***Primary Asset ID*** | ***Primary Asset Name*** | ***Type (information/service)*** | ***Justification*** |
| 0 | *Communications* | service | A communication must be establish between UAS, and to the remote pilot in order to coordinate the swarm and follow commands |
| 1 | *Delivery* | service | An UAS need to know its position to adjust trajectory |
| 2 | *Coordination* | service | Each system need to communicate with the fleet to remain coordinated and to work effectively |
| 3 | *Position / Orientation / Speed* | information | An UAS need to know its orientation to position, orientation and speed to maneuver |
| 4 | *Weather monitoring* | service | The fleet could monitor weather conditions to take decisions avoid damage from the elements |



# Impact Assessment

| 1.2 Impact Assessment on Primary Assets | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| ***Primary Asset Name*** | ***Potential Compromise of C, I or A:*** | ***Impact (see Table in the Methodology)*** | | | | | | | | |
| | | **Personnel** | **Capacity** | **Performance** | **Economic** | **Branding** | **Regulatory** | **Environment** | **Overall Impact (1-2)** | **Justification** |
| Communications | C | 1 | 1 | 1 | 1 | 3 | 2 | 1 | 2 | The leak of information could have some legal consequences and put the company in a bad light |
| | I | 1 | 5 | 5 | 4 | 4 | 2 | 2 | 4 | Integrity ensures the correct functioning of the delivery system |
| | A | 1 | 3 | 4 | 2 | 2 | 2 | 2 | 3 | Availability allows for quick response in case of emergency |
| Delivery | C | 1 | 1 | 1 | 1 | 5 | 5 | 1 | 3 | Consumers data could be at risk |
| | I | 1 | 5 | 1 | 5 | 5 | 3 | 2 | 4 | If integrity is disrupted, the delivery system could be compromised |
| | A | 1 | 5 | 4 | 5 | [illegible] | 3 | 2 | 5 | The unavailablity of the delivery service would compromise the entire company business |
| Coordination | C | 1 | 1 | 1 | 1 | [illegible] | [illegible] | 1 | 1 | If the communication to coordinate the fleet are leaked there are no real consequences |
| | I | 1 | 3 | 5 | 5 | [illegible] | [illegible] | 3 | 4 | UAS could collide with each other |
| | A | 1 | 3 | 5 | 5 | [illegible] | [illegible] | 3 | 4 | UAS could collide with each other |
| Position / Orientation / Speed | C | 1 | 1 | 1 | 1 | [illegible] | 2 | 1 | 1 | If the sensors data are leaked there are no real consequences |
| | I | 1 | 2 | 4 | 4 | 2 | 2 | 4 | 4 | Single UAS could lose control |
| | A | 1 | 2 | 4 | 4 | 2 | 2 | 4 | 4 | Single UAS could lose control |
| Weather monitoring | C | 1 | 1 | 1 | 1 | 3 | 1 | 1 | 2 | Not securing access to weather monitoring can be seen as a weakness in the access system |
| | I | 1 | 3 | 3 | 3 | 2 | 1 | 1 | 3 | Loss of integrity could led the fleet through a storm which could compromise the mission |
| | A | 1 | 3 | 3 | 3 | 2 | 1 | 1 | 3 | Loss of availability could led the fleet through a storm which could compromise the mission |



# Supporting Assets



| ***Primary Asset / Supporting Asset*** | Communications | Delivery | Coordination | Position / Orientation / Speed | Weather monitoring |
|---|---|---|---|---|---|
| Network interfaces | X | | | | X |
| Network protocol | X | | | | X |
| Antennas | X | | | | X |
| GPS | | | X | X | X |
| Accelerometer | | | | X | |
| Database service | | X | | | |
| *Weather sensors* | | | | | X |
| *Lidar system* | | X | X | X | |
| *Propellers* | | X | X | | |
| *Brushless motors* | | X | X | | |
| Battery | X | X | X | X | X |

# Primary Assets

- Power
- Monitoring
- LAN e Internet Connections

| Primary Asset ID | Primary Asset Name | Type (information/service) | Justification |
|---|---|---|---|
| COMM | *Communication with UAS & Data Link* | Service | Enables real-time communication between Unmanned Aircraft Systems (UAS) and the UTM system. A failure could result in loss of control, data corruption, or delays in flight coordination, affecting operational efficiency and safety |
| CONG | *Congestion management* | Service | Manages air traffic density to prevent conflicts and maintain safe separation between UAS. If compromised, there could be flight path overlaps, increased collision risks, and inefficiencies in airspace utilization. |
| TAOA | *Terrain and Obstacle Avoidance:* | Service | Ensures that UAS navigate safely around buildings, mountains, and other obstacles. A failure in this system could lead to collisions, mission failures, or significant safety hazards. |
| RPRR | *Route Planning and Re-routing* | Service | Determines optimal flight paths and enables real-time adjustments due to changing conditions (e.g., weather, congestion). If unavailable, drones may follow outdated or unsafe routes, increasing accident risks. |
| GEOF | *Dynamic Geo fencing surveillance* | Sevice | Establishes virtual boundaries and prevents UAS from entering restricted or hazardous airspace. A breach could lead to security incidents, privacy violations, or non-compliance with aviation regulations. |
| AUTH | *Authentication & Authorization System* | Service | Ensures that only authorized users and drones can access the network and issue commands. A breach could allow malicious actors to take control of drones or manipulate flight plans, leading to security threats. |
| WEWI | *Weather and wind Integration* | Information | Provides real-time and forecasted meteorological data to inform drone operations. Failure in this system could expose UAS to dangerous weather conditions such as strong winds, storms, or turbulence. |
| HIST | Historical data and Flight Logs | Information | Stores flight paths, incident records, and compliance data. If compromised, historical data would be lost or tampered with, making investigations and audits impossible. |
| COOR | *Real time GPS coordinates and flight plans* | Information | Provides precise geolocation and flight trajectory data. If compromised, drones may lose positioning accuracy, leading to navigational errors, unauthorized deviations, or mid-air collisions. |
| PORT | *Portable UTM System Functionality* | Service | Supports operations such as agricultural needs, humanitarian efforts, and other applications |

# Impact Assessment

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| *Communication with UAS & Data Link* | **C** | 2 | 4 | 4 | 3 | 2 | 1 | 1 | 4 | Disrupting communication with UAS affects operations, causing safety risks and regulatory violations. |
| | **I** | 2 | 3 | 5 | 2 | 3 | 4 | 1 | 5 | |
| | **A** | 2 | 4 | 3 | 3 | 1 | 4 | 1 | 4 | |
| *Congestion management* | **C** | 1 | 1 | 3 | 1 | 1 | 1 | 1 | 3 | nability to manage congestion can cause mid-air conflicts, delays, and economic loss. |
| | **I** | 4 | 2 | 4 | 3 | 2 | 3 | 4 | 4 | |
| | **A** | 4 | 3 | 4 | 3 | 2 | 3 | 4 | 4 | |
| *Terrain and Obstacle Avoidance:* | **C** | 1 | 2 | 2 | 4 | 1 | 1 | 2 | 4 | Failure in obstacle detection leads to collisions, potential human casualties, and environmental damage. |
| | **I** | 4 | 4 | 4 | 3 | 3 | 3 | 5 | 5 | |
| | **A** | 4 | 4 | 4 | 4 | 1 | 3 | 5 | 5 | |
| *Route Planning and Re-routing* | **C** | 4 | 4 | [illegible] | [illegible] | [illegible] | 2 | 1 | 4 | Losing dynamic routing causes inefficient airspace use, safety concerns, and financial losses. |
| | **I** | 3 | 4 | [illegible] | [illegible] | [illegible] | 1 | 3 | 4 | |
| | **A** | 3 | 4 | [illegible] | [illegible] | [illegible] | 1 | 1 | 4 | |
| *Dynamic Geo fencing surveillance* | **C** | 2 | 1 | [illegible] | [illegible] | [illegible] | 1 | 1 | 2 | Unauthorized access to geo-fencing data enables security breaches, unauthorized flights, and regulatory violations. |
| | **I** | 2 | 3 | [illegible] | [illegible] | [illegible] | 4 | 2 | 4 | |
| | **A** | 2 | 4 | [illegible] | [illegible] | [illegible] | 4 | 2 | 4 | |
| *Authentication & Authorization System* | **C** | 1 | 2 | [illegible] | [illegible] | [illegible] | 1 | 1 | 5 | Unauthorized access to critical systems compromises security, regulatory compliance, and safety. |
| | **I** | 1 | 1 | [illegible] | [illegible] | [illegible] | 1 | 1 | 4 | |
| | **A** | 1 | 1 | 4 | 5 | 3 | 1 | 1 | 5 | |
| *Weather and wind Integration* | **C** | 1 | 1 | 1 | 2 | 1 | 1 | 1 | 2 | Incorrect weather data affects flight safety, efficiency, and compliance. |
| | **I** | 1 | 4 | 5 | 3 | 1 | 1 | 1 | 5 | |
| | **A** | 1 | 4 | 5 | 3 | 1 | 1 | 1 | 5 | |
| Historical data and Flight Logs | **C** | 1 | 1 | 1 | 3 | 1 | 1 | 1 | 3 | Data leaks expose sensitive operational data, affecting security and trust. |
| | **I** | 1 | 1 | 4 | 2 | 1 | 1 | 1 | 4 | |
| | **A** | 1 | 1 | 3 | 2 | 1 | 1 | 1 | 3 | |
| *Real time GPS coordinates and flight plans* | **C** | 3 | 1 | 2 | 5 | 1 | 1 | 1 | 5 | GPS data manipulation leads to loss of navigation, airspace conflicts, and catastrophic safety risks. |
| | **I** | 4 | 1 | 4 | 2 | 2 | 4 | 1 | 4 | |
| | **A** | 4 | 1 | 4 | 3 | 4 | 4 | 1 | 4 | |
| *Portable UTM System Functionality* | **C** | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | The impossibility to utilize a portable UTM may deter certain users from buying into the system |
| | **I** | 3 | 1 | 3 | 3 | 3 | 1 | 1 | 3 | |
| | **A** | 1 | 1 | 4 | 3 | 3 | 1 | 1 | 4 | |

# Supporting Assets



| ***Primary Asset / Supporting Asset*** | Communication with UAS & Data Link | Congestion management | Terrain and Obstacle Avoidance: | Route Planning and Re-routing | Dynamic Geo fencing surveillance |
|---|---|---|---|---|---|
| *Ground and satellite network infrastructure* | x | x | | | |
| *Air traffic management software* | | x | | x | x |
| *Lidar and Radar sensors* | | | | | |
| *Autenthication software* | x | x | | x | |
| *Human personnel and sourveillance* | | x | | x | x |
| *Servers, data centers* | x | | | | |
| *UAVs* | x | x | | x | |
| *Prediction algorithms* | | x | x | x | |
| | | | | | |
| *Transportable hardware and software packages* | x | | | x | x |
| Alerting systems | | x | x | | |

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# Primary Assets

| Primary Asset ID | Primary Asset Name | Type (information/service) | Justification |
|---|---|---|---|
| 1 | *UTM SYSTEM* | SERVICE | manages airspace design and real-time operations, foundational for airspace management |
| 2 | *COMMUNICATION INFRASTRUCTURE* | SERVICE | ensures secure vehicle-to-vehicle and vehicle-to-service communication, for real-time data exchange and control |
| 3 | *FLIGHT CONTROL SYSTEM* | SERVICE | enables automated routing, automation and coordination, important for fleet autonomy |
| 4 | *AUTHENTICATION DATA* | INFORMATION | protect against unauthorized UAS access |
| 5 | *FLIGHT DATABASE* | INFORMATION | store airspace restriction and trajectories and weather constraints, important for flight planning |
| 6 | *REAL-TIME TELEMETRY DATA* | INFORMATION | live monitoring of UAV position and environmental conditions, important for ensure safety |
| 7 | *EMERGENCY RESPONSE* | SERVICE | handles emergencies (lost link, collisions ...) and recovery protocols |
| 8 | *INTERNET CONNECTIVITY* | SERVICE | allow connection between UTM, UAS and cloud services, important for life weather information and authentication, vulnerable to DDos |
| 9 | *OPERATOR DASHBOARD* | SERVICE | interface for monitoring and intervention, exposed to phising |
| 10 | *ACCESS MANAGEMENT SYSTEM* | SERVICE | maneges user permission and role-based access |

- **Power**
- **Monitoring**

# Impact Assessment (1)

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| UTM SYSTEM | C | 3 | 2 | 2 | 4 | 4 | 4 | 1 | 4 |
| | I | 4 | 4 | 4 | 4 | 3 | 4 | 2 | 4 |
| | A | 4 | 5 | 5 | 5 | 4 | 4 | 2 | 5 |
| COMMUNICATION INFRASTRUCTURE | C | 1 | 1 | 2 | 2 | 2 | 2 | 1 | 2 |
| | I | 2 | 2 | 2 | 2 | 1 | 2 | 1 | 2 |
| | A | 3 | 3 | 4 | 4 | 3 | 3 | 1 | 4 |
| FLIGHT CONTROL SYSTEM | C | 2 | 1 | 1 | 2 | 2 | 3 | 1 | 2 |
| | I | 5 | 4 | 5 | 4 | 3 | 5 | 3 | 5 |
| | A | 5 | 5 | 5 | 5 | 4 | 4 | 3 | 5 |
| AUTHENTICATION DATA | C | 3 | 2 | 2 | 3 | 4 | 4 | 1 | 3 |
| | I | 3 | 3 | 3 | 3 | 4 | 4 | 1 | 4 |
| | A | 3 | 3 | 4 | 4 | 3 | 3 | 1 | 4 |
| FLIGHT DATABASE | C | 2 | 1 | 1 | 2 | 2 | 3 | 1 | 2 |
| | I | 4 | 4 | 4 | 4 | 3 | 4 | 2 | 4 |
| | A | 3 | 4 | 4 | 4 | 3 | 3 | 1 | 4 |
| REAL-TIME TELEMETRY DATA | C | 2 | 1 | 2 | 2 | 2 | 2 | 1 | 2 |
| | I | 4 | 3 | 4 | 3 | 2 | 4 | 2 | 4 |
| | A | 4 | 4 | 5 | 4 | 3 | 3 | 2 | 5 |

# Impact Assessment (2)

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| *EMERGENCY RESPONSE* | C | 2 | 1 | 1 | 2 | 2 | 2 | 1 | 2 |
| | I | 5 | 3 | 4 | 3 | 3 | 4 | 4 | 5 |
| | A | 5 | 4 | 5 | 4 | 4 | 4 | 4 | 5 |
| *INTERNET CONNECTIVITY* | C | 1 | 1 | 2 | 2 | 2 | 2 | 1 | 2 |
| | I | [illegible] | [illegible] | [illegible] | 2 | 1 | 2 | 1 | 2 |
| | A | [illegible] | [illegible] | [illegible] | 4 | 3 | 3 | 1 | 4 |
| *OPERATOR DASHBOARD* | C | [illegible] | [illegible] | [illegible] | 2 | 2 | 2 | 1 | 2 |
| | I | [illegible] | [illegible] | [illegible] | 3 | 2 | 3 | 1 | 3 |
| | A | [illegible] | [illegible] | [illegible] | 4 | 3 | 3 | 1 | 4 |
| *ACCESS MANAGEMENT SYSTEM* | C | 3 | 2 | 2 | 3 | 4 | 4 | 1 | 4 |
| | I | 3 | 3 | 3 | 3 | 4 | 4 | 1 | 4 |
| | A | 3 | 4 | 4 | 4 | 3 | 3 | 1 | 4 |

# Supporting Assets

| ***Primary Asset / Supporting Asset*** | UTM SYSTEM | COMMUNICATION INFRASTRUCTURE | FLIGHT CONTROL SYSTEM | AUTHENTICATION | FLIGHT DATABASE | REAL-TIME TELEMETRY DATA | EMERGENCY RESPONSE | INTERNET CONNECTIVITY | OPERATOR DASHBOARD | ACCESS MANAGEMENT SYSTEM |
|---|---|---|---|---|---|---|---|---|---|---|
| | server cloud | Network routers | GPS sensors | | base | Telemetry sensors | ontingency protoco | Routers | CSS file | Audit tools |
| | database | TCP/IP protocols | Control firmware | ess control list | ems | ansmission protoco | otification module | Firewalls | HTML file | Access policies |
| | APIs | wifi sensor | Routing modules | Digital certificates | Ls | APIs | Backup power | Satellite links | Rendering servers | evocation mechanism |
| Airspace management soft | | cryption modules | cle detection sens | r authentication sy | ftwa | mpression algorit | mergency landing | k redundancy syste | bSocket services | identity providers |
| | | Signal boosters | battery | Security tokens | on too | ime analytics engi | satellite phones | modems | monitors | d access control syste |
| | encryption modules | battery | altimeters | Ide manag | validation tools | ound control statio | ools for restricted | ork monitoring tools | ator training modul | or authentication tools |
| | | Radio transceivers | gyroscopes | | Access control mechanisms | | | | | APIs |
| | | Ground-based relay stations | accelerometers | | | | | | | |
| | | | Navigation software | | | | | | | |
| | | | | | | | | | | |
| | | | | | | | | | | |

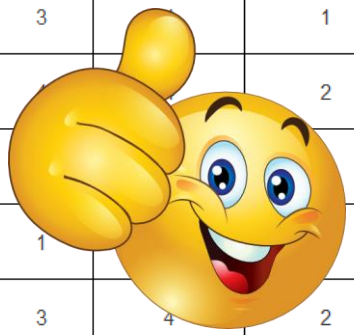
# Primary Assets

| | | | | |
|---|---|---|---|---|
| PA1 | *User account credential* | information | in order to access the workstation | USER DOMAIN |
| PA2 | *Personal user knowledge and experience* | service | to better solve every task | |
| PA3 | *Control of UAS operational licenses* | service | to determine if the UAS operator has the right to carry out their activities | WORKSTATION DOMAIN |
| PA4 | *Identification of the codes for UAS types* | information | to distinct correctly the type of UAS and redirect toward the appropriate route (accept/reject the flight plan) | |
| PA5 | *Organizational communications among team members* | service | fundamental for a proper operational execution of every task | LAN DOMAIN |
| PA6 | *Classification of priority communications* | service | hierarchical classification for a better local coordination | |
| PA7 | *Two-ways flight data communications* | information | to constantly track the drones during their route | LAN TO WAN DOMAIN |
| PA8 | *Evasive directives in case of emergency* | information | to react immediately in case of need (DAA) | |
| PA9 | *Government communications on temporary airspace restrictions* | information | possibly related to demonstrations or special events | WAN DOMAIN |
| PA10 | *Constantly updated meteorological data* | information | to avoid complications related to adverse weather conditions | |
| PA11 | *Territorial mapping of the area of competence* | service | with maps that include both artificial and natural elements of the terrain | SYSTEM/APPLICATION DO |
| PA12 | *Regulations protocols in the UAS routes* | information | to manage efficently the directions in each route | |
| PA13 | *OTP key system* | service | to secure the access to the local network remotely | REMOTE ACCESS DOMAIN |
| PA14 | *Provisions in coordination with other UTM hubs* | information | to carry out operations concerning a broader area that falls under the jurisdiction of multipe UTM | |

- **Power**
- **Monitoring**
- **Internet Connection**

# Impact Assessment

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| PA7 | C | 2 | 2 | 2 | 3 | 3 | 3 | 2 | 3 | this diffusion could allow thieves to intercept and precisely locate the most valuable cargo |
| | I | 2 | 4 | 4 | 3 | 2 | 1 | 1 | 4 | high risk of possible vehicle collisions due to inaccurate information |
| | A | 2 | 3 | 4 | 4 | 3 | 2 | 1 | 4 | if it is not possible to locate the UAS, it becames very difficult/almost impossible to coordinate the routes |
| PA8 | C | 1 | 3 | 2 | 4 | 3 | [illegible] | 1 | 4 | this type of information can be maliciously used against the company to anticipate security measures |
| | I | 3 | 3 | 4 | 4 | [illegible] | [illegible] | 2 | 4 | wrong measures that can harm not only the company but also the environment |
| | A | 2 | 3 | 3 | 3 | [illegible] | [illegible] | [illegible] | 3 | the absence of countermeasure in case of emergencies will lead to company vulnerability |
| PA9 | C | 1 | 1 | 1 | 1 | 1 | [illegible] | [illegible] | 1 | not a real problem if people knows about government restriction |
| | I | 3 | 4 | 4 | 4 | 3 | 4 | 2 | 4 | it could be very dangerous with possible repercussion in legal and public order matters |
| | A | 2 | 3 | 3 | 3 | 3 | 2 | 1 | 3 | if not available it could lead to operative complication |
| PA10 | C | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | it's not a problem if everyone knows this information |
| | I | 3 | 4 | 4 | 4 | 3 | 1 | 2 | 4 | it can lead to severe errors with the route management and possible damage to the fleet |
| | A | 3 | 3 | 3 | 3 | 2 | 1 | 2 | 3 | the absence can lead to operational inefficiency due to programmatic impossibility |



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# Supporting Assets

eit Digital MASTER SCHOOL

| ***Primary Asset / Supporting Asset*** | PA1 | PA2 | PA3 | PA4 | PA5 | PA6 |
|---|---|---|---|---|---|---|
| *Internet connection* | x | | | | x | |
| *Academic and professional training (bachelor/master)* | | x | x | x | x | x |
| *List of authorized certificate types* | | | x | | | |
| *List of UAS codes* | | | | x | | |
| *Internal telephone switchboard* | | | | | x | x |
| *Internal hierarchical roles* | x | x | | | x | x |
| *Signal receivers/antennas* | | | | x | | |
| *Alarm system* | | | | | | |
| *Corporate certified email* | x | | | | x | x |
| *Humidity sensors* | | | | | | |
| *Satellite images* | | | | | | |
| *Low altitude airspace aviation code* | | | | x | | |
| *OTP key generator application* | x | | | | | |
| *Cloud-based virtual meeting rooms* | | | | | x | |

**Vrae?**
Afrikaans

**Questions?**
English

**¿Preguntas?**
Spanish

**Cwestiynau?**
Welsh

**Imibuzo?**
Zulu

**Quaestiones?**
Latin

**Ερωτήσεις;**
Greek

**Вопросы?**
Russian

**Fragen?**
German

**問題呢?**
Chinese

**Domande?**
Italian

**أسئلة؟**
Arabic

**質問？**
Japanese

**שאלות?**
Jewish

**tupoQghachmey?**
Klingon

**Questions?**
French